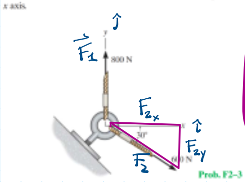
F2–3. Determine the magnitude of the resultant force and its direction measured counterclockwise from the positive x axis.



Prob. F2–3

Not this approach

$F_1 = 800$, $600 = F_2$, $30°$, F_R

Soh Cah Toa

$\sin\theta = \frac{opp.}{hyp}$

$\cos\theta = \frac{adj}{hyp}$

$\tan\theta = \frac{opp.}{adj.}$

hyp. opp adj θ

$\vec{F}_1 = 0\hat{\imath} + 800\hat{\jmath}$ N

$\vec{F}_2 = 519.62\hat{\imath} - 300\hat{\jmath}$ N

$\cos 30° = \frac{F_{2x}}{F_2}$

$F_{2x} = F_2 \cdot \cos(30°) = 600\cos(30°) = 600 \cdot \frac{\sqrt{3}}{2} = 519.62$

$\sin 30° = \frac{F_{2y}}{F_2}$

$F_{2y} = F_2 \sin(30°) = 600 \cdot \sin(30°) = 600\left(\frac{1}{2}\right) = 300$

$\vec{F}_R = (0 + 519.62)\hat{\imath} + (800 - 300)\hat{\jmath}$ N

$\vec{F}_R = 519.62\hat{\imath} + 500\hat{\jmath}$ N

F_R, α, F_{2y}, F_{1y}, $F_{Ry} = 500$, F_{1x}, F_{2x}

$F_{Rx} = 519.62$

$F_R = 721.1$ N

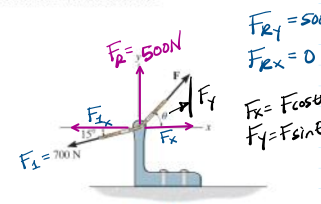
$F_R^2 = F_{Rx}^2 + F_{Ry}^2$

$F_R = \sqrt{(519.62)^2 + (500)^2}$

$\tan\alpha = \frac{F_{Ry}}{F_{Rx}} \rightarrow \alpha = \tan^{-1}\left(\frac{F_{Ry}}{F_{Rx}}\right)$

$\alpha = \tan^{-1}\left(\frac{500}{519.62}\right) = 43.9°$

2–2. If the magnitude of the resultant force is to be 500 N, directed along the positive y axis, determine the magnitude of force **F** and its direction θ.



Probs. 2–1/2

$F_{Ry} = 500N$

$F_{Rx} = 0$

$F_x = F\cos\theta$

$F_y = F\sin\theta$

* Closed system of Equations

2 unknowns, need 2 equations

① $\underbrace{F\cos(\theta)}_{F_x} = \underbrace{700\cos(15°)}_{F_{1x}}$

$F_x + F_{1x} = F_{Rx}$

$F\cos\theta - 700\cos(15) = 0$

② $F\sin\theta - 700\sin(15°) = 500$

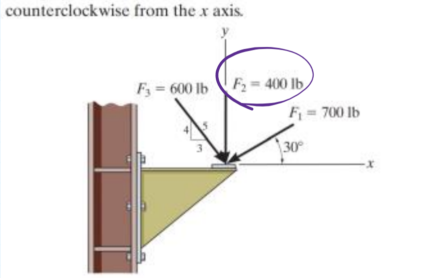
F_{1x}, F_{1y}, 15°, 700

$F_{1x} = 700 \cdot \cos(15°)$

$F_{1y} = 700\sin(15°)$

F2–9. Determine the magnitude of the resultant force acting on the corbel and its direction θ measured counterclockwise from the x axis.

Replace with single force



Prob. F2–9

F_3 600 lb

$F_{3y} = -480$ lb

$F_{3x} = 360$ lb

F_1 700 lb

$F_{1y} = 700\sin(30°) = -350$ lb

F_{1y}, θ, F_R, F_2, F_{Ry}, F_{3y}, F_{1x}, F_{3x}, F_{Rx}

$F_{Ry} = 350 + 400 + 480$

$F_{Ry} = 1230$ lb ↓

$F_{Rx} = 606.2 - 360$

$F_{Rx} = 246.2$ ←

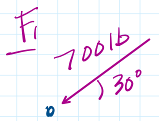
$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}$

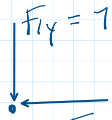
1254.4 lb 11.3°

1254.4° 78.7°

$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}$

$\angle 78.7$





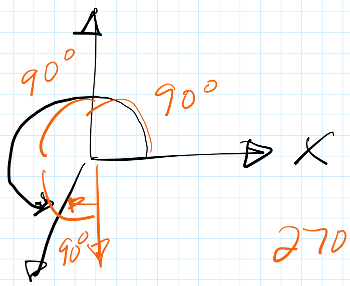
$F_{1y} = 700 \sin(30°) = -350\,lb$

$F_{1x} = 700 \cos(30°) = -606.2\,lb$

$F_R = 1254.4\ lb$

$\tan\theta = \dfrac{246.2}{1230}$

$\theta = \tan^{-1}\left(\dfrac{246.2}{1230}\right) = 11.3°$



$270° - 11.3°$

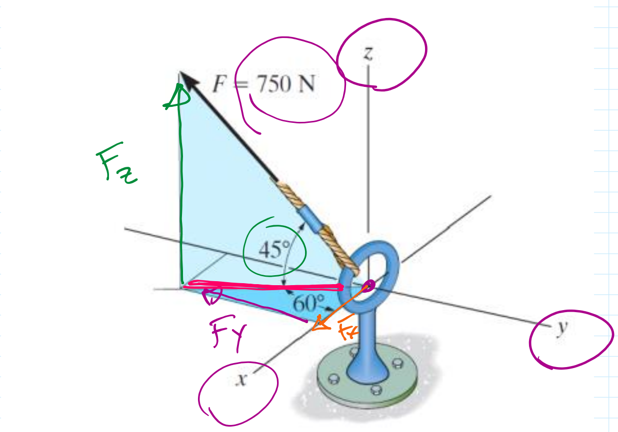
$258.7°$

$F = m \cdot a$

$1\ N = 1\ kg \cdot 1\ m/s^2$

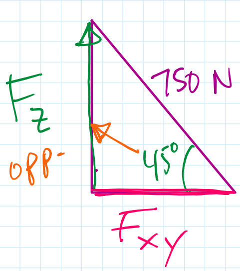
$1\ lb = 1\ slug \cdot 1\ ft/s^2$

F2–17. Express the force as a Cartesian vector.



Prob. F2–17

$\vec{F} = \underline{265.2}\ \hat{i} - \underline{459.3}\ \hat{j} + \underline{530.3}\ \hat{k}\ [N]$

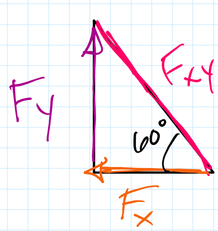


$F_z = 750 \sin(45°)$

$F_z = 530.3\ N$

$F_{xy} = 750 \cos(45°)$

$F_{xy} = 530.3 N$



$F_x = F_{xy} \cos(60°)$

$F_x = 530.3 \cos(60°) = 265.2 N$

$F_y = F_{xy} \sin(60°)$

$F_y = 530.3 \sin(60°) = -459.3 N$